

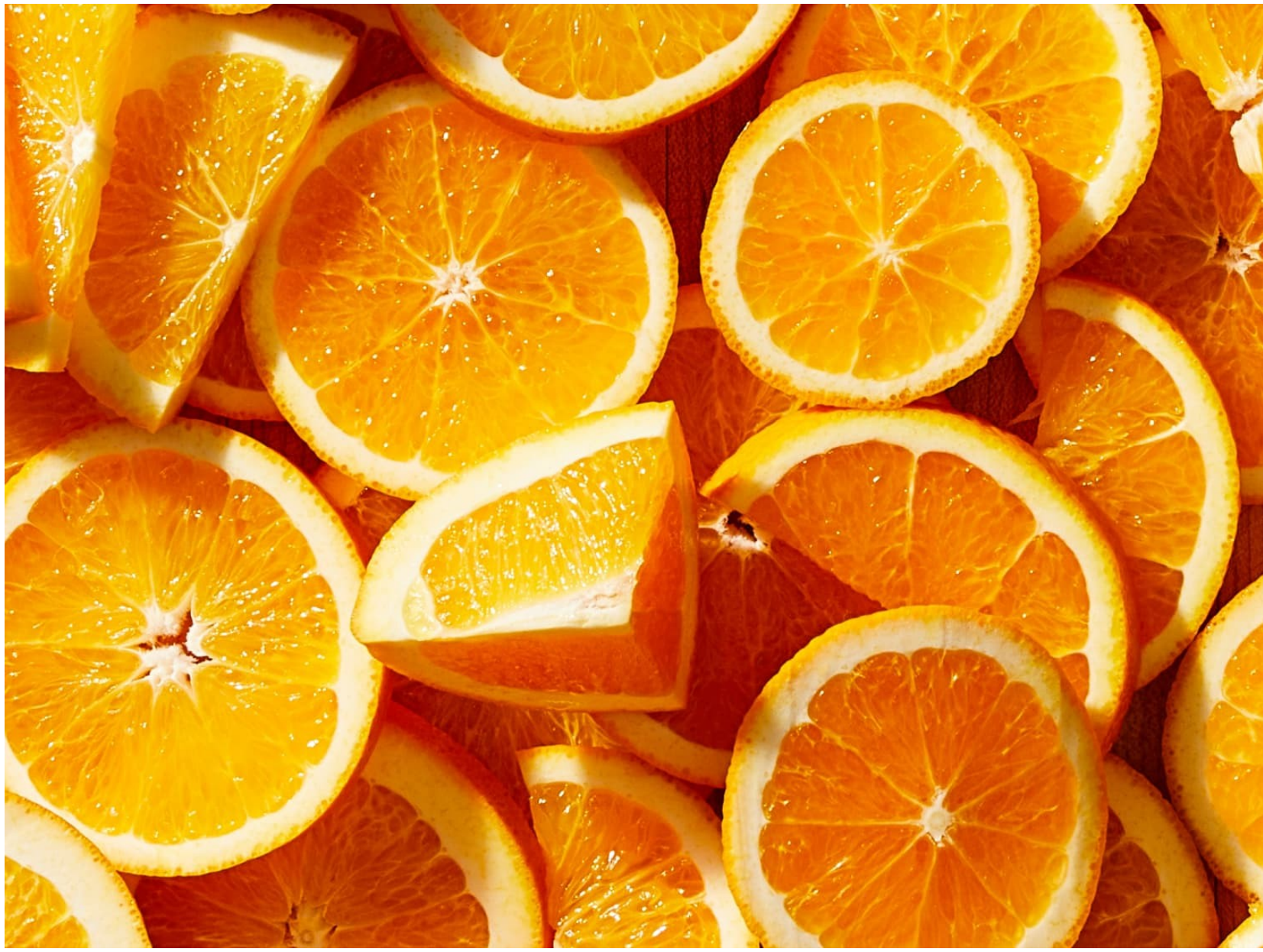
BCOG 100

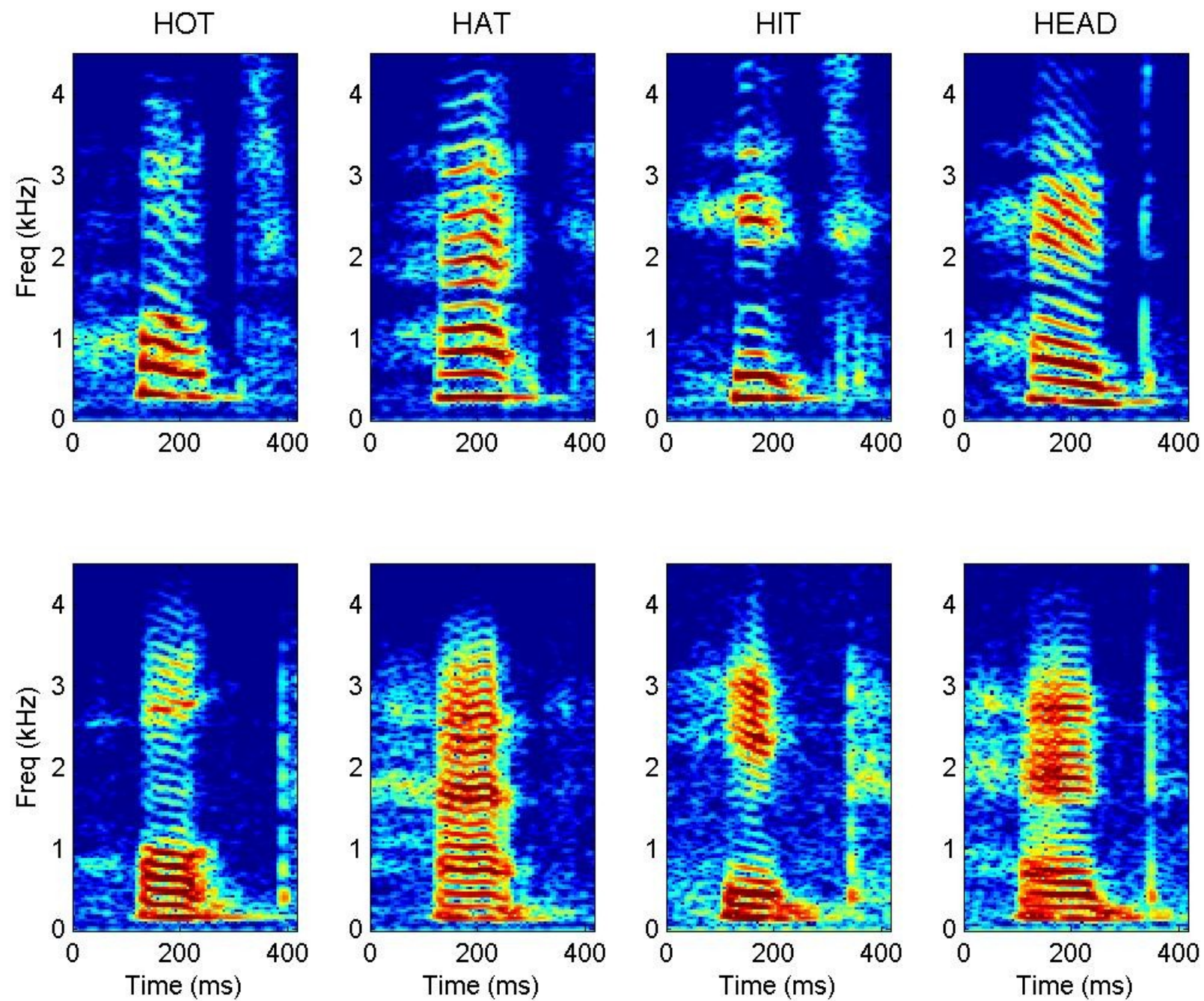
Lecture 1.0: Course Overview



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN













Our Schedule

- Week 1: BCS Overview
- Week 2: Philosophy of Mind (NO CLASS Mon Sep 1 and Fri Sep 5)
- Week 3: Philosophy of Mind
- Week 4: Neurons and Information Processing
- Week 5: Neural Circuits
- Week 6: Brains
- Week 7: Associative and Reinforcement Learning
- Week 8: Memory and Generative Models
- Week 9: Social Cognition
- Week 10: Language and Symbolic Thought
- Week 11: Cognitive Psychology and Cognitive Neuroscience
- Week 12: Machine Learning & AI
- Week 13: Large Language Models (NO CLASS Fri Dec 12)
- Week 13: FALL BREAK
- Week 14: Nature vs. Nurture 1
- Week 15: Nature vs. Nurture 2 (NO CLASS Fri 12/12)



What is Brain and Cognitive Science?

The study of the mind and the brain, with two notable properties:

- Interdisciplinary
- Computational



Interdisciplinary BCS

Psychology
Neuroscience
Philosophy
Computer Science and Artificial Intelligence
Anthropology
Linguistics
Education
Ecology and Evolutionary Biology



Computational BCS

Computational Methods

- Computer-based experiments
- Computer-driven tools like EEG & fMRI
- Complex data analysis
- Computational models and simulations

Computational Metaphor

- Is the mind or brain like a computer?
- What do we gain by treating it as such?
- What do we lose?

